

Analysis Workflow

Brent Oil Change Point Analysis | Birhan Energies | Task 1

1. Objective

Quantify how major geopolitical events, OPEC decisions, and economic shocks are associated with structural shifts in Brent crude oil prices (daily, 20-May-1987 to 14-Nov-2022), and communicate the findings to investors, policymakers, and energy companies.

2. Planned Analysis Steps

- Data loading and cleaning** - parse data/BrentOilPrices.csv, standardise the mixed date formats, sort chronologically, check for gaps/duplicates.
- Exploratory data analysis** - plot the raw price series; compute and plot log returns $\log(\text{price}_t) - \log(\text{price}_{t-1})$; test stationarity (ADF, KPSS) on both series; examine volatility clustering via rolling standard deviation. (*Completed* - see notebooks/eda.ipynb and reports/interim_report.pdf.)
- Event research and compilation** - compile a structured list of 15+ major events (wars, sanctions, OPEC decisions, financial crises) with approximate start dates into data/events.csv. (*Completed*.)
- Bayesian change point modeling (Task 2)** - build a PyMC model with a discrete-uniform prior over the switch point τ , separate "before" and "after" mean parameters (μ_1, μ_2), a `pm.math.switch` function linking τ to the active mean, and a Normal likelihood. Run MCMC via `pm.sample()`.
- Convergence and posterior diagnostics** - check $\hat{r} \sim 1.0$ via `pm.summary()`, inspect trace plots (`pm.plot_trace()`), and plot the posterior distribution of τ to assess how sharply the change point is localised in time.
- Event association** - map the posterior mode of τ (and any additional change points from extended/multiple-change-point models) to the nearest entries in data/events.csv, and formulate hypotheses about which event(s) plausibly triggered each detected shift.
- Impact quantification** - for each associated change point, report the before/after posterior means, the absolute and percentage change, and a probabilistic statement (e.g. " $P(\mu_2 > \mu_1) = 0.98$ ").
- Dashboard (Task 3)** - expose the price series, change point results, and event correlations via a Flask API, and build a React frontend with filtering, event highlighting, and drill-down.
- Reporting** - write the final report (blog-post format) with embedded visualisations, quantified impact statements, limitations, and future work.

3. Communication Channels

Audience	Channel / Format
Internal team / tutors	Slack #all-week10, GitHub issues and project board for task tracking
Investors / analysts	Interactive dashboard (Task 3) for self-service exploration of price vs. events

Policymakers / government bodies	Written report (PDF / Medium-style blog post) with plain-language, quantified impact statements and visualisations
Energy companies (operational planning)	Summary tables of historical regime shifts and their magnitude/duration, delivered alongside the dashboard

4. References

See the project README.md for the full reading list (Bayesian change point detection, PyMC documentation, MCMC background) used to inform the modeling approach in Task 2.